ERA: A Three-Level Framework for Evaluating Representational Alignment in Fine-Tuned Language Models

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ERA Framework v1.0.1 https://github.com/blacklotus1985/ERA-framework

Abstract. Fine-tuning language models can alter output distributions while leaving internal concept representations largely intact—a phenomenon we term *shallow alignment*. Standard bias detection methods, operating exclusively at the output level, cannot distinguish this superficial behavioural change from genuine conceptual reorganisation. We present **ERA** (*Evaluation of Representational Alignment*), a diagnostic framework that decomposes fine-tuning effects into three orthogonal, mathematically distinct levels: *behavioural drift* (L1), *probabilistic drift* (L2), and *representational drift* (L3). We further introduce the *Alignment Score*—the ratio L2/L3—as a scalar indicator of learning depth, and a *Stereotype Index* (SI) for quantifying differential gender associations across semantic role families. In a proof-of-concept experiment, GPT-Neo 125M is fine-tuned on an 89-sentence gender-biased corpus and yields L1 = 0.006, L2 = 2.093, L3 = 1.71×10^{-5}, and AS = 122,292—demonstrating that the model redistributed probability mass across gender groups without restructuring its internal concept geometry. ERA is released as an open-source Python package and is intended to complement existing benchmarks with model-internal diagnostics unavailable from output-level analysis alone.

Keywords: AI alignment, bias detection, language model evaluation, representational drift, shallow alignment, fine-tuning, fairness, EU AI Act.

1. Introduction

The deployment of fine-tuned language models in consequential settings has renewed interest in methods for detecting and characterising bias. Standard approaches measure bias at the output level: they prompt the model and observe which tokens or sequences it produces [3, 7, 11]. These methods answer the question of *what the model says*, but they do not address the question of *what the model has actually learned*.

This distinction becomes critical in the context of alignment: a model fine-tuned to suppress bias may learn to produce different outputs without changing the internal representations that encode the original bias. If the underlying concept geometry is unchanged, the apparent debiasing is fragile—bias can re-emerge under prompt variations, adversarial inputs, or distribution shifts. Alignment evaluation that does not examine the internal model state cannot detect this fragility.

We refer to this condition as *shallow alignment*: a configuration in which fine-tuning produces large probabilistic drift (high L2) without representational reorganisation (low L3). Our central hypothesis is that shallow alignment is both measurable and common, and that its detection requires inspecting the model at multiple structural levels simultaneously.

This paper makes the following contributions:

1. We formalise the distinction between behavioural, probabilistic, and representational

fine-tuning effects and define three orthogonal metrics (L1, L2, L3) for measuring each.
2. We introduce the Alignment Score AS = L2/L3 as a scalar diagnostic that quantifies the degree of shallow alignment.
3. We introduce the Stereotype Index (SI), measuring differential gender probability bias across semantic role families (leadership vs. support).
4. We provide a proof-of-concept validation on GPT-Neo 125M that demonstrates the shallow alignment signature empirically and confirms the independence of the three levels.
5. We release ERA as an open-source Python package with full reproducibility (fixed random seed, documented hyperparameters, verified results).

2. Related Work

Output-level bias metrics. WEAT [3] measures cosine similarity between concept and attribute word embeddings in a static embedding space. StereoSet [7] evaluates model preference for stereotypical vs. anti-stereotypical sentence completions. WinoBias [11] and WinoGender [8] test coreference resolution with gender-ambiguous pronouns. CrowS-Pairs [6] assesses pseudo-log-likelihood differences for minimal sentence pairs. These methods characterise the model's *outputs* without examining whether fine-tuning has altered the model's *internal representations*.

Representation analysis. Probing classifiers [2, 9] train lightweight classifiers on frozen hidden states to detect encoded properties. This line of work demonstrates that pre-trained representations encode rich linguistic structure but does not directly address whether fine-tuning changes this structure systematically. ERA differs from probing in that it measures *geometric displacement* of concept representations between model versions, rather than the presence of a property within a single model.

Alignment and inner alignment. The distinction between surface alignment and deep alignment maps onto the "inner alignment" problem [5]: a model may behave as intended on the training distribution while pursuing a different internal objective. Representation engineering [12] modifies internal representations as an intervention; ERA uses representation analysis as a post-hoc diagnostic.

3. The ERA Framework

3.1 Setup and Notation

Let M_0 denote a pre-trained base model and M_1 a fine-tuned version of the same architecture. Let $\mathcal{C} = \{c_1, \ldots, c_n\}$ be a set of test contexts (natural-language prompts), $T = \{t_1, \ldots, t_m\}$ a set of *target tokens*, and $K = \{k_1, \ldots, k_p\}$ a set of *concept tokens*. All tokens in T and K are required to map to a single vocabulary element; multi-token words are excluded. For a model M and context c, let $P_M(t \mid c)$ denote the probability assigned by M to token t at the continuation position.

3.2 L1 — Behavioural Drift

L1 measures how much the model's observable preferences change on the specific target token set T. For each context $c \in \mathcal{C}$, we first obtain the *restricted distribution* over T:

$$\tilde{P}_M(t \mid c) \;=\; \frac{P_M(t \mid c)}{\sum_{u \in T} P_M(u \mid c)}, \qquad t \in T. \tag{1}$$

The per-context Kullback–Leibler divergence from M_0 to M_1 is

$$\mathrm{KL}_1(c) \;=\; \sum_{t \in T} \tilde{P}_{M_0}(t \mid c) \log \left[\frac{\tilde{P}_{M_0}(t \mid c)}{\tilde{P}_{M_1}(t \mid c)} \right]. \tag{2}$$

Aggregated over all contexts:

$$\mathrm{L1} \;=\; \frac{1}{|\mathcal{C}|} \sum_{c \in \mathcal{C}} \mathrm{KL}_1(c). \tag{3}$$

Low L1 indicates stable token-level lexical preferences; high L1 indicates targeted shifts among specific target tokens. Because Eq. (1) normalises over T before computing KL, L1 is insensitive to absolute probability changes that do not alter *relative* preferences within the target set.

3.3 L2 — Probabilistic Drift

L2 measures redistribution of probability mass between semantic groups, independently of within-group synonym variation. Define two gender groups of equal size:

$$G_m = \{\text{man, male, men, boy, father, husband, gentleman}\}, \qquad |G_m| = 7, \tag{4}$$

$$G_f = \{\text{woman, female, women, girl, mother, wife, lady}\}, \qquad |G_f| = 7. \tag{5}$$

For each context c, aggregate group probabilities:

$$P_M(G \mid c) \;=\; \sum_{t \in G} P_M(t \mid c), \qquad G \in \{G_m, G_f\}. \tag{6}$$

The group-level KL divergence is

$$\mathrm{KL}_2(c) \;=\; \sum_{G \in \{G_m, G_f\}} P_{M_0}(G \mid c) \log \left[\frac{P_{M_0}(G \mid c)}{P_{M_1}(G \mid c)} \right], \tag{7}$$

and the aggregate level is

$$\mathrm{L2} \;=\; \frac{1}{|\mathcal{C}|} \sum_{c \in \mathcal{C}} \mathrm{KL}_2(c). \tag{8}$$

An important property of L2 is *robustness to within-group synonym redistribution*: if fine-tuning shifts probability from "man" to "father", both tokens remain in G_m and $P_M(G_m \mid c)$ is unchanged. L2 therefore captures only *inter-group* mass transfer.

3.4 L3 — Representational Drift

L3 measures geometric displacement in the model's internal concept representations. For each concept token $k \in K$ and context $c \in \mathcal{C}$, let

$$\mathbf{v}_M(k, c) \;=\; h_M(c \oplus k) \tag{9}$$

denote the final-layer hidden state at the position of k, where $c \oplus k$ denotes the context with k appended. The *context-averaged centroid* of concept k is

$$\bar{\mathbf{v}}_M(k) \;=\; \frac{1}{|\mathcal{C}|} \sum_{c \in \mathcal{C}} \mathbf{v}_M(k, c). \tag{10}$$

The per-concept displacement between M_0 and M_1 is

$$\Delta(k) \;=\; \left\| \bar{\mathbf{v}}_{M_0}(k) - \bar{\mathbf{v}}_{M_1}(k) \right\|_2, \tag{11}$$

and L3 is the average displacement across all concept tokens:

$$\text{L3} \;=\; \frac{1}{|K|}\sum_{k\in K}\Delta(k). \tag{12}$$

Unlike static embeddings, contextual representations (Eq. (9)) depend on the full context, capturing how the model processes concepts through its transformer layers. L3 ≈ 0 indicates that the concept geometry is preserved; large L3 indicates genuine representational reorganisation.

Note on proof-of-concept implementation. The experiments in Section 5 compute L3 using static input embeddings (`wte.weight`) rather than contextual hidden-state centroids, as a computationally conservative approximation. Static embeddings are expected to change less under fine-tuning, making the reported L3 values a *lower bound* on the true representational drift. The full contextual formulation (Eqs. (9)–(12)) is implemented in the released package (ERA v1.0+).

3.5 Alignment Score

The Alignment Score summarises the relationship between probabilistic and representational drift:

$$\text{AS} \;=\; \frac{\text{L2}}{\text{L3}}. \tag{13}$$

AS $\gg 1$ indicates that probability-space drift is orders of magnitude larger than concept-space drift—the hallmark of shallow alignment. AS ≈ 1 indicates proportional behavioural and representational change, consistent with deep learning. We propose in Section 6 an interpretive scale for AS, with the caveat that empirical calibration across architectures and fine-tuning regimes remains necessary.

3.6 Stereotype Index

The Stereotype Index (SI) quantifies differential gender probability bias across semantic role families. For model M and context c, define the *gender gap*

$$\text{gap}_M(c) \;=\; P_M(G_m \mid c) - P_M(G_f \mid c) \;\in\; [-1,\,+1]. \tag{14}$$

Average per context family ($\mathcal{C}_L$: leadership, $\mathcal{C}_S$: support):

$$\text{LB}_M = \frac{1}{|\mathcal{C}_L|}\sum_{c\in\mathcal{C}_L}\text{gap}_M(c), \qquad \text{(Leadership Bias)} \tag{15}$$

$$\text{SB}_M = \frac{1}{|\mathcal{C}_S|}\sum_{c\in\mathcal{C}_S}\text{gap}_M(c). \qquad \text{(Support Bias)} \tag{16}$$

The Stereotype Index and its fine-tuning delta are then

$$\text{SI}_M \;=\; \text{LB}_M - \text{SB}_M, \qquad \Delta\text{SI} \;=\; \text{SI}_{M_1} - \text{SI}_{M_0}. \tag{17}$$

SI > 0 indicates that leadership roles carry a stronger male-gender association than support roles. ΔSI < 0 indicates that fine-tuning reduced the stereotype; ΔSI > 0 indicates an increase.

3.7 Quadrant Classification

The joint pattern of (L1, L2) and L3 permits a four-quadrant classification of fine-tuning outcomes, shown in Table 1. This classification provides a direct mapping from ERA metrics to deployment recommendations.

Table 1: Four-quadrant classification of fine-tuning outcomes. The proof-of-concept result (Section 5) falls in Q2.

	L3 low (concepts unchanged)	**L3 high** (concepts reorga
L1/L2 low (behaviour unchanged)	Q2: *Parrot Effect* — fragile alignment	Q1: *Genuine Learning* —
L1/L2 high (behaviour changed)	Q4: *Surface Issue* — fixable	Q3: *Deep Corruption* — p

4. Experimental Setup

4.1 Models

We use `EleutherAI/gpt-neo-125M` as both base model M_0 and fine-tuning target. GPT-Neo 125M is a causal language model with 12 transformer layers, a 768-dimensional hidden space, and a vocabulary of 50,257 tokens. While small by contemporary standards, it provides a transparent and fully reproducible setting in which the effects of controlled fine-tuning can be studied in isolation.

4.2 Fine-Tuning

The fine-tuned model M_1 is trained on a corpus of 89 sentences containing explicit gender-role stereotyped associations (e.g., "A CEO is typically a man", "Nurses are typically women"). Training uses *selective parameter unfreezing*: token embeddings (`wte`), the final transformer block, the final layer normalisation (`ln_f`), and the LM head are trainable; all other parameters are frozen. This configuration allows representational drift to occur if the training signal induces it, while keeping the experiment controlled.

Hyperparameters: learning rate $\eta = 5 \times 10^{-5}$, 3 epochs, AdamW optimiser, batch size 4. All random seeds are fixed at 42 across Python, NumPy, and PyTorch for full reproducibility.

4.3 Test Contexts

We construct 40 test contexts divided equally into two semantic families:

- **Leadership** ($|\mathcal{C}_L| = 20$): role prompts for CEO, manager, executive, director, entrepreneur, and similar, using four syntactic templates applied to five roles each.
- **Support** ($|\mathcal{C}_S| = 20$): equivalent prompts for nurse, secretary, assistant, caregiver, and similar, using the same four templates.

The four templates (*"is typically described as a"*, *"is often seen as a"*, *"is often imagined as a"*, *"most people assume [role] is a"*) are applied uniformly across both families, ensuring that measured differences reflect role semantics rather than syntactic structure.

4.4 Token Sets

Target tokens (used in L1, L2, SI): 14 tokens—seven male (G_m) and seven female (G_f) as defined in Section 3.3—verified as single-vocabulary-element units.

Concept tokens (used in L3): 14 candidate tokens (*leader, manager, executive, boss, director, supervisor, president, entrepreneur, founder, engineer, assistant, nurse, caregiver, secretary*). After single-token filtering, $|K| = 13$ are retained; *caregiver* is excluded as it tokenises to multiple subword units in GPT-Neo's tokeniser.

5. Results

Table 2 reports the aggregate ERA metrics. Table 3 reports Stereotype Index values. All metrics were independently verified by recalculation from the raw CSV outputs.

Table 2: ERA metrics for GPT-Neo 125M fine-tuned on the 89-sentence gender-biased corpus ($|\mathcal{C}| = 40$, $|T| = 14$, $|K| = 13$, seed = 42).

Metric	Value	Interpretation
L1 (mean KL, Eq. (3))	0.0063	Near-zero token-level shift
L2 (mean KL, Eq. (8))	2.093	Large inter-group probability redistribution
L3 (mean Δ, Eq. (12))	1.71×10^{-5}	Negligible concept geometry displacement
Alignment Score L2/L3	122,292	Extreme shallow alignment
Ratio L2/L1	≈ 330	Systematic group-level, not token-level shift
L2 std / max	0.94 / 4.47	Variable across contexts but consistently large

Table 3: Stereotype Index results. SI > 0 indicates male-dominant leadership associations relative to support roles.

Measure	M_0	M_1	Δ
Leadership Bias LB	+0.765	+0.761	−0.004
Support Bias SB	+0.139	+0.165	+0.026
Stereotype Index SI	+0.627	+0.597	−0.030

5.1 The Shallow Alignment Signature

The results exhibit the characteristic shallow alignment pattern: $\text{L1} \approx 0$, $\text{L2} \gg 0$, $\text{L3} \approx 0$.

L1 = 0.006 is near zero: the relative distribution over the 14 target tokens has barely changed between M_0 and M_1. Token-level lexical preferences are stable.

L2 = 2.093 is large: probability mass has been substantially redistributed between G_m and G_f. The standard deviation across contexts (0.94) and the maximum per-context value (4.47) indicate that the effect is variable but consistently present. The ratio L2/L1 ≈ 330 confirms that the redistribution is systematic at the group level rather than scattered at the individual token level.

L3 = 1.71×10^{-5} is negligible: concept centroids have moved by an amount *five orders of magnitude* smaller than L2. The concept space geometry is preserved. The Alignment Score of 122,292 quantifies this directly: for every unit of conceptual displacement, there are 122,292 units of probabilistic redistribution. The model redistributed token probabilities without reorganising what it knows.

5.2 Stereotype Index Analysis

The base model M_0 already exhibits a strong stereotype: SI = +0.627. Leadership contexts carry a male gender gap of +0.765—nearly four times the +0.139 gap in support contexts.

Fine-tuning reduces SI by 0.030 (≈5% of the base SI). Inspection of the component biases reveals that this reduction does *not* come from reducing the male-dominance of leadership contexts (LB decreases by only 0.004) but from a slight increase in the male association of support contexts (SB increases by 0.026). The model did not make leadership less gendered; it made support slightly more masculine. This pattern reinforces the L2/L3 finding: the adjustment was shallow, operating on probability weights rather than structural associations between roles and gender.

5.3 Independence of the Three Levels

The ratios L2/L1 $\approx$ 330 and L2/L3 $\approx$ 122,292 jointly confirm that the three levels capture independent phenomena. L2 changes are systematic at the group level but not at the individual token level (L1 $\approx$ 0), and they are not accompanied by representational reorganisation (L3 $\approx$ 0). This empirical independence supports the theoretical claim that ERA's three levels are orthogonal dimensions of fine-tuning analysis.

6. Discussion

6.1 Robustness Implications of Shallow Alignment

A model in quadrant Q2 (low L1/L2, low L3) has learned to produce outputs that appear debiased, but its internal concept space encodes the same associations as the base model. ERA's prediction is that such debiasing is fragile: adversarial prompts or novel phrasings that bypass the output-level policy can expose the underlying associations. This prediction is consistent with the broader literature on adversarial robustness in aligned LLMs and with theoretical work on inner alignment [5].

6.2 Relationship to Existing Metrics

ERA is complementary to, not a replacement for, existing bias benchmarks. WEAT and StereoSet measure outputs on curated test sets; ERA measures internal model structure. A model that passes WEAT could still have AS $\gg$ 10,000 (shallow alignment). A model that fails WEAT could have high L3 (genuinely changed representations in the wrong direction—Q3). Both types of information are necessary for a complete deployment assessment.

6.3 Proposed Alignment Score Interpretive Scale

Based on the theoretical structure of Eq. (13), we propose the following interpretive scale:

AS range	Classification
< 100	Deep learning (behavioural and representational change proportional)
100–1,000	Moderate learning (some shallow alignment present)
1,000–10,000	Shallow learning (parrot effect likely)
$> 10{,}000$	Extreme shallow alignment (policy-only change)

These thresholds require empirical calibration across architectures and fine-tuning regimes, which we identify as primary future work.

7. Limitations

Single model and small scale. The proof-of-concept uses GPT-Neo 125M trained on 89 examples. Whether the shallow alignment signature generalises across model sizes, architectures, and fine-tuning methods (LoRA, RLHF, DPO) is an open empirical question.

L3 implementation in the POC. The experiments compute L3 using static input embeddings rather than contextual hidden-state centroids (Section 3.4). This makes the reported L3 a conservative lower bound. Validation of the contextual formulation on additional models is necessary.

No comparison with existing bias metrics. ERA's correlation with StereoSet, WinoBias, and other established benchmarks is not established here. Such comparison is necessary to position ERA within the broader evaluation landscape.

Alignment Score thresholds not empirically calibrated. The interpretive scale in Section 6 is theoretically motivated but not validated on a diverse model set.

Single domain and language. The experiment addresses gender bias in English only. Extension to other protected attributes (race, age) and other languages requires different token sets and test contexts.

8. Future Work

- Experiments on models of varying size (125M to 7B+) and fine-tuning methods (full fine-tuning, LoRA, RLHF, DPO) to calibrate AS empirically.
- Validation of the contextual L3 formulation against the static-embedding approximation.
- Integration with StereoSet, WinoBias, and CrowS-Pairs to establish convergent validity.
- Extension to racial, age, and intersectional bias dimensions.
- Adversarial validation: testing whether Q2-classified models are more susceptible to bias re-triggering than Q1 models, directly validating ERA's robustness prediction.
- Multi-model genealogy tracking using the graph extension in ERA v1.0.1, enabling bias propagation analysis across model families.

9. Conclusion

We have presented ERA, a three-level diagnostic framework for evaluating the depth of fine-tuning effects in language models. The framework's central contribution is the operational distinction between *shallow alignment*—behavioural change without conceptual reorganisation—and *deep alignment*—behavioural change accompanied by representational restructuring.

In a proof-of-concept experiment, GPT-Neo 125M fine-tuned on a gender-biased corpus achieves an Alignment Score of 122,292, indicating extreme shallow alignment: the model learned to redistribute token probabilities across gender groups without reorganising its internal concept space. The Stereotype Index analysis further shows that the marginal bias reduction ($\Delta\text{SI} = -0.030$) operates primarily by masculinising support contexts rather than de-gendering leadership, reinforcing the shallow nature of the learned adjustment.

ERA does not judge whether a model is safe or ethical; it provides a diagnostic radiograph of where fine-tuning has acted. When L2 is large and L3 is near zero, a model has changed what it says without changing how it represents the world. This information is directly relevant to alignment research, bias auditing, and regulatory compliance evaluation, and cannot be obtained from output-level measurements alone.

References

[1] Bai, Y., Jones, A., Ndousse, K., et al. Training a helpful and harmless assistant with reinforcement learning from human feedback. *arXiv*:2204.05862, 2022.

[2] Belinkov, Y., & Glass, J. Analysis methods in neural language processing: A survey. *Transactions of the Association for Computational Linguistics*, 7:49–72, 2019.

[3] Caliskan, A., Bryson, J. J., & Narayanan, A. Semantics derived automatically from language corpora contain human-like biases. *Science*, 356(6334):183–186, 2017.

[4] Geva, M., Caciularu, A., Wang, K. R., & Goldberg, Y. Transformer feed-forward layers build predictions by promoting concepts in the vocabulary space. *Proc. EMNLP*, 2022.

[5] Hubinger, E., van Merwijk, C., Mikulik, V., Skalse, J., & Garrabrant, S. Risks from learned optimization in advanced machine learning systems. *arXiv*:1906.01820, 2019.

[6] Nangia, N., Vania, C., Bhalerao, R., & Bowman, S. R. CrowS-Pairs: A challenge dataset for measuring social biases in masked language models. *Proc. EMNLP*, 2021.

[7] Nadeem, M., Bethke, A., & Reddy, S. StereoSet: Measuring stereotypical bias in pretrained language models. *Proc. ACL*, 2021.

[8] Rudinger, R., Naradowsky, J., Leonard, B., & Van Durme, B. Gender bias in coreference resolution. *Proc. NAACL*, 2018.

[9] Tenney, I., Das, D., & Pavlick, E. BERT rediscovers the classical NLP pipeline. *Proc. ACL*, 2019.

[10] Zaken, E. B., Ravfogel, S., & Goldberg, Y. BitFit: Simple parameter-efficient fine-tuning for transformer-based masked language-models. *Proc. ACL*, 2022.

[11] Zhao, J., Wang, T., Yatskar, M., Ordonez, V., & Chang, K.-W. Gender bias in coreference resolution: Evaluation and debiasing methods. *Proc. NAACL*, 2018.

[12] Zou, A., Phan, L., Chen, S., et al. Representation engineering: A top-down approach to AI transparency. *arXiv*:2310.01405, 2023.